

Three ways to build the same data product

Simple | Solution 2 | Solution 3

From rapid rehearsal to governed, multi-engine medallion delivery

1 FASTEST PATH

2 METADATA-DRIVEN

3 MULTI-ENGINE

**Validated in Hackathon
Demo**

Architecture and decision guide

One outcome, three levels of engineering

SIMPLE

Manual + notebook

Speed

- Fastest to start
- Fewest moving parts
- Best for rehearsal and exploration

Hours

SOLUTION 2

Control-driven ingestion

Repeatability

- Incremental Delta merge
- Reusable metadata pattern
- Isolated Solution2 schema

Days

SOLUTION 3

Multi-engine medallion

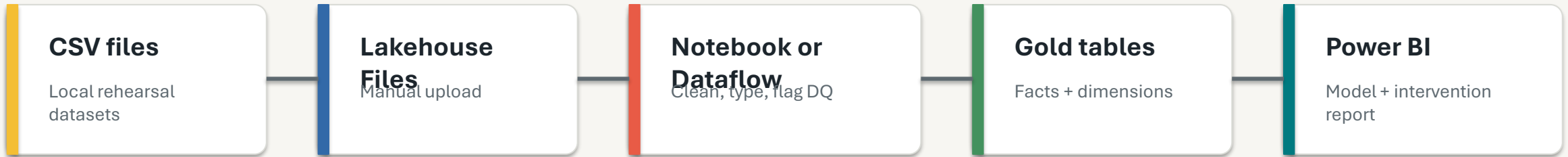
Governance

- Dedicated Bronze/Silver/Gold
- Pipeline + Dataflow + T-SQL
- Warehouse audit and DQ evidence

Production pattern

Simple: quickest route from CSV to insight

BEST FOR: REHEARSAL / PROTOTYPE



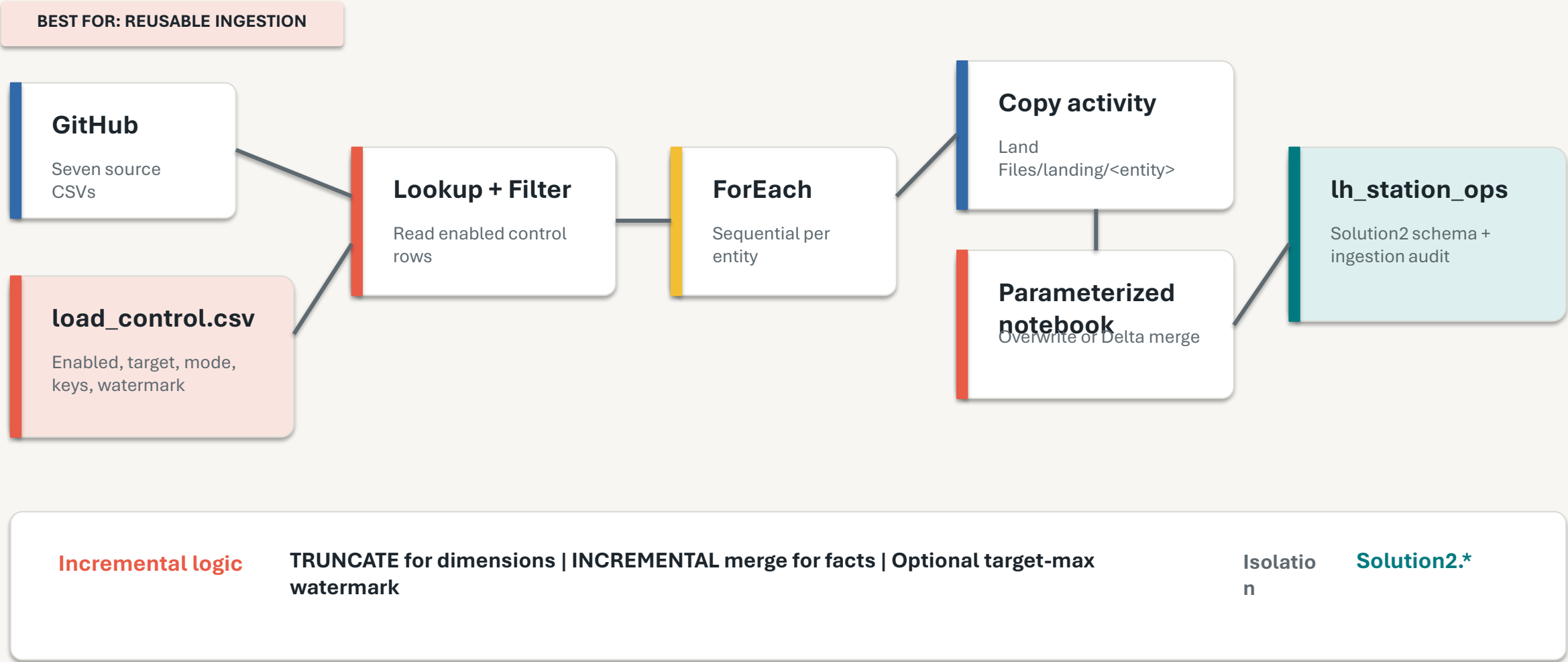
What it optimizes

- Low setup effort and rapid iteration
- Easy inspection in a single notebook
- Direct path to the stakeholder report

Trade-offs

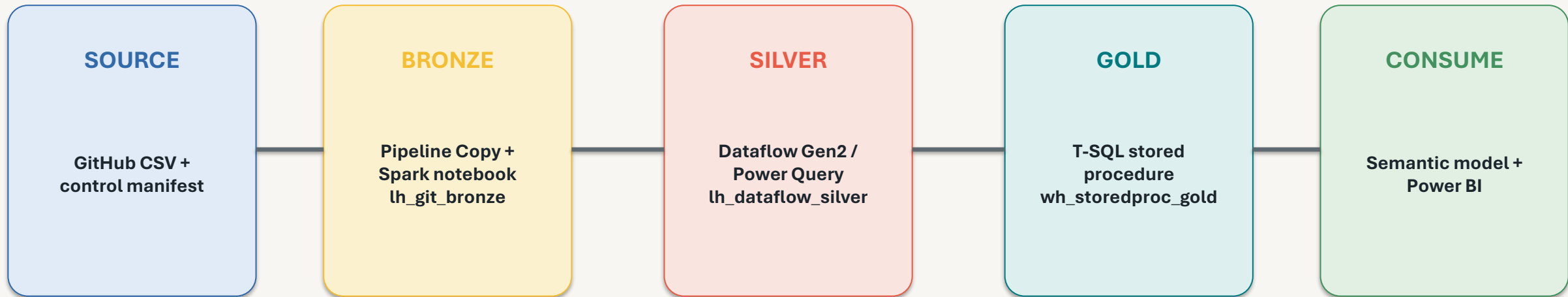
- Manual landing and rerun discipline
- Limited operational audit trail
- Harder to scale across many entities

Metadata-driven incremental ingestion



Multi-engine medallion with operational evidence

BEST FOR: GOVERNED DEMO / PRODUCTION PATTERN



WAREHOUSE OPERATIONS PLANE

pipeline_run

pipeline_step

reconciliation

data quality

Bad records remain visible in Silver and are excluded from Gold

Choose the engineering level that matches the goal

Dimension	Simple	Solution 2	Solution 3
Primary goal	Speed	Repeatable ingest	Governed delivery
Landing	Manual upload	Dynamic Pipeline Copy	Dynamic Pipeline Copy
Transform	Notebook / Dataflow	Parameterized Spark	Spark + Dataflow + T-SQL
Incremental	Manual rerun	Keys + watermark merge	Bronze merge; deterministic Silver/Gold
Isolation	Chosen Lakehouse	Solution2 schema	Dedicated items per layer
Monitoring	Notebook output	Ingestion audit	Run, step, reconcile, DQ tables
Best fit	Workshop	Reusable ingestion	Teaching + production pattern

Solution 3 completed end to end

Validated master run

c87ee117-8bd0-4c77-9dd5-220669ece567

COMPLETED

MASTER PIPELINE STATUS

7 / 7

PIPELINE STEPS COMPLETED

7 / 7

RECONCILIATIONS BALANCED

169

EXPECTED REJECTED ROWS

Data quality findings

- 166 reversed assistance timestamps
- 1 fulfilled count above request count
- 1 negative passenger count
- 1 missing station reference

Design principle

A successful run is not enough.

Prove row balance, quality outcomes, and traceability for every entity.

Start simple, add controls when the risk demands them

Need an answer today?

Use Simple

Manual landing + notebook

Loading many files repeatedly?

Use Solution 2

Control table + incremental merge

Need governance and operational proof?

Use Solution 3

Medallion + audit + DQ

The three approaches are complementary: use them to teach progression, not to declare one universal winner.